

Experiment – 06

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Subject: Advanced Database
Management System

Question 1:

A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges. Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as: Difference = Apples Sold - Oranges Sold Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

Sales

sale_date	fruit	sold_num
2020-05-01	apples	10
2020-05-01	oranges	8
2020-05-02	apples	15
2020-05-02	oranges	15
2020-05-03	apples	20
2020-05-03	oranges	0
2020-05-04	apples	15
2020-05-04	oranges	16

Output:

sale_date	diff
2020-05-01	2
2020-05-02	0
2020-05-03	20

Solution:

```
SELECT
  SALE_DATE,
  SUM(
    CASE
      WHEN FRUIT = 'apples' THEN SOLD_NUM
      ELSE -SOLD_NUM
    END
  ) AS DIFF
FROM SALES AS S
GROUP BY SALE_DATE
ORDER BY SALE_DATE;
```

Question 2:

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where: • id represents the current node. • p_id represents the parent node. Classify every node into one of the following categories: •

Root: A node that does not have any parent. • Leaf: A node that has a parent but does not have any children. • Inner: A node that has both a parent and at least one child. Display the node ID along with its category. Return the result sorted by node ID.

Tree	
id	p_id
1	NULL
2	1
3	1
4	2
5	2

Solution:

```
SELECT
  ID,
  CASE
    WHEN P_ID IS NULL THEN 'ROOT'
    WHEN ID NOT IN (
      SELECT P_ID
      FROM TREE
      WHERE P_ID IS NOT NULL
    ) THEN 'LEAF'
    ELSE 'INNER'
  END AS TYPE
FROM TREE
ORDER BY ID;
```